Relational Transition

SONY

CLIENT: SONY **(**SYNTHETIC)

TEAM NAME: Sony Database Advising Group

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Client: Sony Group Corporation

Data Management Problem:

Sony Group Corporation's current data system consists of organizing customers' information and accounts, warranty, deliveries, service requests with possibilities of replacement and repairs, warranties, and purchases of products.

Requirements:

A customer must create at least one account with Sony, but they can create more than one account with them. When a customer purchases a product, the purchase is recorded using proof of purchase with a receipt number that contains the details of the date of purchase and the purchasing location.

Each product could have a warranty attached to it that contains a start date, date of warranty expiration, status, and the type of coverage the product has. If an issue is presented to a customer, they will submit a service request, which contains the ticket number, a description of the issue, product identification, and the date of the request.

There are two possibilities when a service request is submitted: the product is going to either be replaced or repaired. If a customer uses the repair service the request is sent to the service center, and replacement parts are used, a part identification is detected, name of the part, category, a description of the part, and what is the availability, then a service report is created containing the report identification number, stating if the issue was resolved, location, the SKU of the item and parts used to perform the service, delivery hubs are used to used to deliver to customers where they are identifiable by an identification number, their address, number of transporters and the type of the delivery that was made. Additionally, a customer may contact a Client Service Consultant (employee) who will then submit a service request for either a repair or replacement.

A warehouse, which is identified by a warehouse identification number, the address for the specific warehouse, and its contact information, delivers to a store that is recognized by an identification number, the store name, the address, the type of store, and its contact information. Additionally, when an item is being repaired, replacement parts are stored in the warehouse.

Service centers are where services are provided (items being repaired, etc.) to customers, where they are recognized by their identification number, the addresses where is located, the number of employees, their contact information, and the item capacity for the specific center. The service centers in order to help customers repair items.

Sony customers can process a return ticket that contains the date of return of the product, the reason of return and the condition of the item. After the customer is done with their service on a product a service review may be submitted on their end containing a review identification number, the rating in quality of service, comments and media.

Employees at Sony can be technicians or a client service assistant, employees get an identification number attached to their first and last name, social security number (SSN), date of birth and age that is obtained by their date of birth.

Assumptions:

- One person may have more than 1 account, but at least one must be recorded.
- When a customer sends an item for repair or replacement, it will be sent back
- Assume that there are many more roles at Sony that are not mentioned.
- Assume delivery is on a national scale
- Assume delivery type is a different type (ground or air transport)
- Assume warehouse is working 24/7
- Assume each product can have up to one warranty
- Assume online purchase can be put in the purchase location

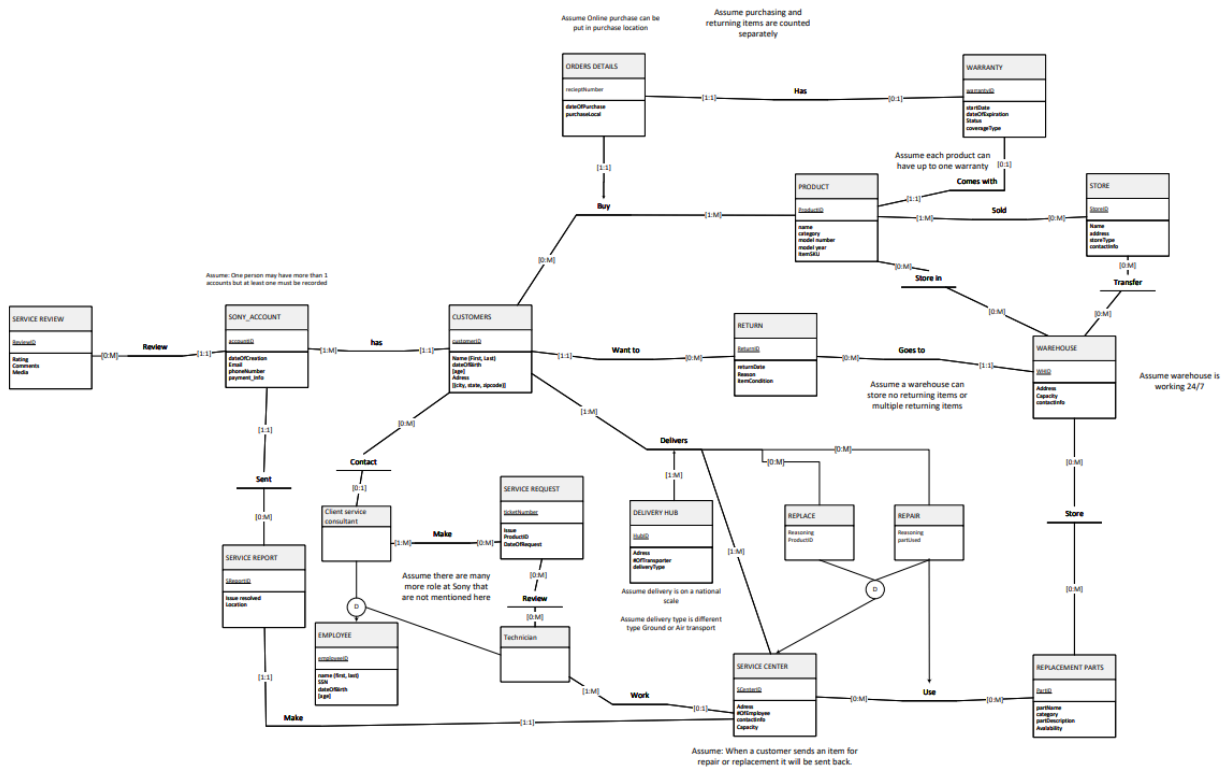
Cardinalities:

- One customer has at least one account with Sony [1:M]
- Customers can purchase at least one Sony or product or more [1:M]

- Each customer who buys a product may have a warranty for that product [0:1]
- A proof of purchase comes with a warranty [1:1]
- Customer can submit and request zero or multiple service requests for Sony products [1:M]
- Customers' request may lead to a repair or a replacement of the product [0:M}
- Delivery hubs may petition a service request or multiple [1:M]
- Customers can request multiple returns [0:M]
- Warehouses can store many replacement parts [0:M]
- Service centers work with at least one technician [1:M]
- Technicians review many service request [0:M]
- Sony accounts may send service reports [0:M]
- Service centers may use many different replacement parts [0:M]
- Delivery hubs deliver to at least one customer [1:M]
- Customers may contact many client service consultants [0:M]
- Client service consultants make at least one service request [1:M]
- Customers may have many proof of purchase [0:M]

Conceptual Schema (ER Diagram)

Figure 1 – ER diagram



ER Data Dictionary Alphabetical list of:

Entity Classes:

Schema Construct	Construct Meaning	Structure/Constraints
Customers	Entity Class, models customer data	
● Customer ID	Identifying Attribute	
● Name (first, last)	Composite attribute	Not NULL
● dateOfBirth	Customer DOB	
● age	Derived from dateOfBirth	
● Address (city, state, zipcode)	Composite attribute	Not NULL
DELIVERY HUB	Entity Class, models shipment hubs	
● HubID	Identifying Attribute	
● address	Location	Not NULL

● NumberofTransporter	Number of transport vehicles	More than 1
● deliveryType	Type of shipment	(ground, air)
EMPLOYEE	Entity Class, models Sony employees	
● employeeID	Identifying Attribute	
● name (first, last)	Composite attribute	Not NULL
● SSN	Unique employee SSN	Unique
● dateOfBirth	Employee DOB	
● age	Derived attribute	
CLIENT SERVICE CONSULTANT	Sub-Class of EMPLOYEE	
TECHNICIAN	Sub-Class of EMPLOYEE	
PRODUCT	Entity Class, models Sony products	

● ProductID	Identifying Attribute	
● name	Product name	Not NULL
● category	Product category	
● modelNumber	Unique model number	Unique
● modelYear	Year released	More than 2000
ORDER DETAILS	Entity Class, models purchase transactions	
● receiptNumber	Identifying Attribute	Unique
● dateOfPurchase	Purchase date	Not NULL
● purchaseLocal	Store or Online	Optional
REPLACEMENT PARTS	Entity Class, models repair parts	
● PartID	Identifying Attribute	
● partName	Name of part	Not NULL

● category	Part Category	
● partDescription	Description	
● Availability	Inventory status	(in stock, or out of stock)
RETURN	Weak Entity, models product returns	Depends on CUSTOMER & PRODUCT
● returnDate	Date returned	
● Reason	Reason for return	
● itemCondition	Condition upon return	
SERVICE CENTER	Entity Class, models repair facilities	
● SCenterID	Identifying Attribute	
● address	Location	
● NumderofEmployee	Number of staff	

● contactInfo	Contact information	More than 1
● Capacity	Max capacity for Service	
REPAIR	Sub-Class of SERVICE_CENTER	
● Reasoning	Repair reason	
● partUsed	References REPLACEMENT_PARTS	
REPLACE	Sub-Class of SERVICE_CENTER	
● Reasoning	Replacement reason	
● ProductID	References PRODUCT	
SERVICE REPORT	Entity Class, models service results	
● SReportID	Identifying Attribute	PK
● IssueResolved	Resolution description	

● Location	Service location	
SERVICE REQUEST	Weak Entity	
● ticketNumber	Identifying Attribute	PK
● Issue	Problem description	Not NULL
● ProductID	Product References	
● DateOfRequest	Date ticket was created	Not NULL
SERVICE REVIEW	Entity Class, models customer feedback	
● ReviewID	Identifying Attribute	
● Rating	Service rating	
● Comments	Written feedback	
● Media	Optional attachment	
SONY ACCOUNT	Entity Class, models online accounts	

● accountID	Identifying Attribute	PK
● dateOfCreation	Account creation date	
● Email	Unique email	Not NULL
● phoneNumber	Contact number	
● paymentInfo	Payment details	
STORE	Entity Class, models retail stores	
● StoreID	Identifying Attribute	PK
● Name	Store name	Not NULL
● address	Location	
● storeType	Store classification	
● contactInfo	Contact info	
WARRANTY	Entity Class, models product warranty	

● warrantyID	Identifying Attribute	PK
● startDate	Start date	
● dateOfExpiration	Expiration date	
● Status	Warranty status	
● CoverageType	Coverage category	
WAREHOUSE	Entity Class, models storage facilities	PK
● WHID	Identifying Attribute	
● Address	Warehouse address	
● Capacity	Storage capacity	
● contactInfo	Contact info	

Attributes (list from Alphabetical Order)

Attribute	Construct Meaning	Constraints / Notes
accountID (PK)	Unique identifier for Sony account	Primary Key
Address	Physical location address	
age (derived)	Derived from date of birth	Derived
Availability	Inventory status of part	In stock / Out of stock
Capacity	Maximum storage or service capacity	
category	Classification of product or part	
Comments	Customer written feedback	Optional
contactInfo	Contact information	More than 1
coverageType	Warranty coverage category	
customerID (PK)	Unique identifier for customer	Primary Key
dateOfBirth	Date of birth	
dateOfCreation	Account creation date	
dateOfExpiration	Warranty expiration date	

dateOfPurchase	Date product was purchased	Not NULL
DateOfRequest	Date service ticket was created	Not NULL
deliveryType	Type of delivery service	Ground / Air
Email	Customer email address	Unique, Not NULL
employeeID (PK)	Unique employee identifier	Primary Key
HubID (PK)	Unique delivery hub identifier	Primary Key
Issue	Problem description	Not NULL
IssueResolved	Resolution description	
ItemCondition	Condition of product upon return	
Location	Service or facility location	
Media	Optional attachment (image/video)	Optional
modelNumber	Unique product model number	Unique
modelYear	Year product released	More than 2000
Name (first, last)	Full name of person	Composite, Not NULL
partDescription	Description of replacement part	

partName	Name of replacement part	Not NULL
PartID	Unique replacement part identifier	Primary Key
partUsed	Replacement parts used in service	Foreign Key
paymentInfo	Customer payment details	
phoneNumber	Contact phone number	
ProductID	Unique product identifier	Primary Key
purchaseLocal	Purchase location (store or online)	Optional
Rating	Service quality rating	
receiptNumber	Unique purchase receipt identifier	Primary Key
Reason	Reason for product return	
Reasoning	Reason for repair or replacement	
returnDate	Date product was returned	
ReviewID	Unique service review identifier	Primary Key
SCenterID	Unique service center identifier	Primary Key
SReportID	Unique service report identifier	Primary Key

SSN	Employee social security number	Unique
startDate	Warranty start date	
Status	Warranty status	
StoreID	Unique retail store identifier	Primary Key
storeType	Classification of store	
ticketNumber	Unique service request ticket	Primary Key
warrantyID	Unique warranty identifier	Primary Key
WHID	Unique warehouse identifier	Primary Key
Number Of Employee	Total number of employees	
Number Of Transporter	Total number of transport vehicles	More than 1

RELATIONSHIP (Listed in Alphabetical Order)

Schema	Construct Meaning	Structure/Constraints
BUY	Customer purchases product	Each purchases are 1 product
COMES WITH	Product has warranty	Each product has 1 warranty
CONTACT	Customer contacts consultant	Optional
DELIVERS	Shipping to the Customer	(Ground, Air)
HAS	Customer Ows account	Has more than 1 account
MAY REQUEST	Customer Creates Ticket	Customer created many tickets
RETURNS	Customer returns product	Valid purchase
REVIEW	Customer writes review	Optional
SOLD	Product sold in store	Many
STORE	Warehouse stores parts	Many

STORE IN	Products are stored in Warehouse	Many
USE	Service center uses parts	Many
WORK	The Technicians are working at Service Center	Many Technicians

Constraints that are NOT shown in Diagram

- Account Rule
  - Each Customer MUST have at least 1 account
- Warehouse
  - The Warehouse Availability is constant
- Return
  - Return is only possible when there is a valid PROOF

Explanation of Abbreviations

- PK: *Primary Key*
- Weak Entity: *This entity depends on another entity for identification*

Translated Relations (Class Notation)

CUSTOMERS(customerID, firstName, lastName, dateOfBirth)

CUSTOMERADDRESS(customerID, zipcode city, state)

- *Foreign Key CUSTOMERADDRESS(customerID) references CUSTOMERS(customerID)*

CUSTOMERDELIVERYPOINT*(customerID, hubID)*

- *Foreign Key CUSTOMERDELIVERYPOINT(customerID) references CUSTOMERS(customerID)*
- *Foreign Key CUSTOMERDELIVERYPOINT(hubID) references DELIVERYHUB(hubID)*

DELIVERYHUB(hubID, hubAddress, #OfTransporter, deliveryType)

- *Foreign Key DELIVERYHUB(customerID) references CUSTOMERS(customerID)*

EMPLOYEES(employeeID, firstName, lastName, SSN, dateOfBirth)

ClientServiceConsultant(employeeID)

- *Foreign Key ClientServiceConsultant(employeeID) references EMPLOYEES(employeeID)*

CONSULTANTREQUEST*(employeeID, ticketNumber)*

- *Foreign Key EMPLOYEEREQUEST(employeeID) references ClientServiceConsultant(employeeID)*
- *Foreign Key EMPLOYEEREQUEST(ticketNumber) references SERVICEREQUEST(ticketNumber)*

Technician(employeeID)

- *Foreign Key Technician(employeeID) references EMPLOYEES(employeeID)*

WORKINGDETAILS(employeeID, SCenterID)

- *Foreign Key WORKINGDETAILS(employeeID) references Technician(employeeID)*
- *Foreign Key WORKINGDETAILS(SCenterID) references SERVICECENTER(SCenterID)*

TECHREVIEW(employeeID, ticketNumber)

- *Foreign Key TECHREVIEW(employeeID) references Technician(employeeID)*
- *Foreign Key TECHREVIEW(ticketNumber) references SERVICEREQUEST(ticketNumber)*

ORDERDETAILS(receiptNumber, warrantyID, productID, storeID, customerID, dateOfPurchase, purchaseLocal)

- *Foreign Key ORDERDETAILS(warrantyID) references WARRANTY(warrantyID)*
- *Foreign Key ORDERDETAILS(storeID) references STORE(storeID)*
- *Foreign Key ORDERDETAILS(customerID) references CUSTOMERS(customerID)*

- *Foreign Key ORDERDETAILS(productID) references PRODUCT(productID)*

PRODUCT(productID, name, category, modelNumber, modelYear, itemSKU)

REPLACEMENTPARTS(partID, partName, category, partDescription, availability)

WAREHOUSESTORAGE(partID, WHID)

- *Foreign Key WAREHOUSESTORAGE(WHID) references WAREHOUSE(WHID)*
- *Foreign Key WAREHOUSESTORAGE(partID) references REPLACEMENTPARTS(partID)*

RETURN(returnID, customerID, WHID, returnDate, reason, itemCondition)

- *Foreign Key RETURN(customerID) references CUSTOMERS(customerID)*
- Foreign Key RETURN(WHID) references WAREHOUSE(WHID)

SERVICECENTER(SCenterID, serviceAddress, #OfEmployee, contactInfo, capacity)

REPLACE(SCenterID, productID, reasoning)

- *Foreign Key REPLACE(SCenterID) references SERVICECENTER(SCenterID)*
- *Foreign Key REPLACE(productID) references PRODUCT(productID)*

REPAIR(SCenterID, reasoning)

- *Foreign Key REPAIR(SCenterID) references SERVICECENTER(SCenterID)*
  REPAIRPART*(SCenterID, partID)*
- *Foreign Key REPAIRPART(SCenterID) references REPAIR(SCenterID)*
- *Foreign Key REPAIRPART(partID) references REPLACEMENTPARTS(partID)*

DELIVERYSERVICE(SCenterID, hubID)

- *Foreign key DELIVERYSERVICE(hubID) references DELIVERYHUB(hubID)*
- *Foreign key DELIVERYSERVICE(SCenterID) references SERVICECENTER(SCenterID)*

SERVICEREPORT(SReportID, SCenterID, partID, issueResolved, location)

- *Foreign Key SERVICEREPORT(SCenterID) references SERVICECENTER(SCenterID)*
- *Foreign Key SERVICEREPORT(partID) references REPLACEMENTPARTS(partID)*

SERVICEREVIEW(reviewID, accountID, rating, comments, media)

- *Foreign Key SERVICEREVIEW(accountID) references SONYACCOUNT(accountID)*

SERVICEREQUEST (ticketnumber, customerID, issue, productID, dateOfRequest)

- Foreign key SERVICE REQUEST (productID) references PRODUCT (productID)
- *Foreign Key SERVICEREQUEST(customerID) references CUSTOMERS(customerID)*

SONYACCOUNT(accountID, customerID, dateOfCreation, Email, phoneNumber, paymentInfo)

- *Foreign Key SONYACCOUNT(customerID) references CUSTOMERS(customerID)*

STORE(storeID, name, address, storeType, contactInfo)

WAREHOUSESTORE(WHID, storeID)

- *Foreign Key WAREHOUSESTORE(WHID) references WAREHOUSE(WHID)*
- *Foreign Key WAREHOUSESTORE(storeID) references STORE(storeID)*

WAREHOUSE(WHID, warehouseAddress, capacity, contactInfo)

WARRANTY(warrantyID, productID, startDate, dateOfExpiration, Status, coverageType)

- *Foreign Key WARRANTY(productID) references PRODUCT(productID)*

Data Dictionary:

Data Dictionary (Relational) - Alphabetical
Customers
Relation for the entity class CUSTOMERS

Attribute	*Data Type*	*Constraint / Notes*
Customer ID	*CHAR(10)*	*Primary Key. Format: C followed by 9 digits (CHECK). NOT NULL*
FirstName	*VARCHAR(30)*	*NOT NULL*
LastName	*VARCHAR(30)*	*NOT NULL*
DateOfBirth	*DATE*	*NULLABLE; CHECK (DateOfBirth <= CURRENT_DATE)*
Email	*VARCHAR(100)*	*UNIQUE: NOT NULL*
PhoneNumber	*VARCHAR(15)*	*NULLABLE; CHECK (phone format)*
Payment_Info	*VARCHAR(200)*	*NULLABLE; encrypted at application layer*
Address_City	*VARCHAR(50)*	*NULLABLE*
Adress_State	*VARCHAR(50)*	*NULLABLE*
Address_ZipCode	*VARCHAR(10)*	*NULLABLE*
SonyAccountID	*CHAR(10)*	*FK (nullable) references SONY_ACCOUNT (AccountID); note: business rule requires each customer to have ≥1 account (enforce via application or periodic DB check)*

DELIVERY_HUB
Relation for the entity class DELIVERY_HUB

Attribute	*Data Type*	*Constraint / Notes*
HubID	*CHAR(8)*	*Primary Key; NOT NULL*
Address	*VARCHAR(200)*	*NOT NULL*
#OfTransporter	*NUMBER(4)*	*NOT NULL; CHECK (#OfTransporter >= 0)*
DeliveryType	*VARCHAR(10)*	*NOT NULL;CHECK (DeliveryType IN ('Ground','Air'))*
ContactInfo	*VARCHAR(100)*	*NULLABLE*
LocationNotes	*VARCHAR(200)*	*NULLABLE*

EMPLOYEE
Relation for the superclass EMPLOYEE

Attribute	*Data Type*	*Constraint / Notes*
EmployeeID	*CHAR(10)*	*Primary Key; NOT NULL*
FirstName	*VARCHAR(30)*	*NOT NULL*
LastName	*VARCHAR(30)*	*NOT NULL*
SSN	*CHAR(11)*	*UNIQUE, NULLABLE; store encrypted*
DateOfBirth	*DATE*	*NULLABLE; CHECK (DateOfBirth <= CURRENT_DATE)*
RoleType	*VARCHAR(30)*	*NOT NULL; values include*

		'Technician', 'ClientServiceConsultant', etc.
ServiceCenterID	*CHAR(8)*	*FK (nullable) references SERVICE_CENTER(Service CenterID)*
ContactInfo	*VARCHAR(100)*	*NULLABLE*

TECHNICIAN
Relation for subclass TECHNICIAN (indented under EMPLOYEE)

Attribute	*Data Type*	*Constraint / Notes*
EmployeeID	*CHAR(10)*	*Primary Key; FK references EMPLOYEE (EmployeeID) (1:1)*
Certifications	*VARCHAR(200)*	*NULLABLE*
Specialization	*VARCHAR(100)*	*NULLABLE*

CLIENT_SERVICE_CONSULTANT
Relation for subclass CLIENT_SERVICE_CONSULTANT (indented under EMPLOYEE)

Attribute	*Data Type*	*Constraint / Notes*
EmployeeID	*CHAR(10)*	*Primary Key; FK references EMPLOYEE (EmployeeID) (1:1)*
Shift	*VARCHAR(20)*	*NULLABLE*
ContactExtension	*VARCHAR(10)*	*NULLABLE*

ORDER_DETAILS
Relation for the entity/relationship ORDER_DETAILS (orders and purchase details)

Attribute	*Data Type*	*Constraint / Notes*
OrderDetailID	*CHAR(12)*	*Primary Key; NOT NULL*
Receipt Number	*VARCHAR(30)*	*UNIQUE; NOT NULL*
CustomerID	*CHAR(10)*	*FK references CUSTOMERS (CustomerID); NOT NULL*
ProductID	*CHAR(12)*	*FK references PRODUCT (ProductID); NOT NULL*
DateOfPurchase	*DATE*	*NOT NULL*
PurchaseLocation	*VARCHAR(100)*	*NULLABLE (store address or 'Online')*
Quantity	*NUMBER(6)*	*NOT NULL; CHECK (Quantity > 0)*
UnitPrice	*NUMBER(12,2)*	*NOT NULL; CHECK (UnitPrice >=0)*

PRODUCT
Relation for the entity class PRODUCT

Attribute	*Data Type*	*Constraint / Notes*
ProductID	*CHAR(12)*	*Primary Key; NOT NULL*
ItemSKU	*VARCHAR(30)*	*UNIQUE; NOT NULL*
Name	*VARCHAR(100)*	*NOT NULL*
Category	*VARCHAR(50)*	*NULLABLE*

ModelYear	*NUMBER(4)*	*NULLABLE; CHECK (ModelYear BETWEEN 1900 AND EXTRACT(YEAR FROM CURRENT_DATE))*
Description	*VARCHAR(500)*	*NULLABLE*
StoreID	*CHAR(8)*	*FK (nullable) references STORE(StoreID)*
WarehouseID	*CHAR(8)*	*FK (nullable) references WAREHOUSE (WarehouseID)*
Price	*NUMBER(12,2)*	*NOT NULL; CHECK (Price >=0)*
WarrantyID	*CHAR(10)*	*FK (nullable) references WARRANTY(WarrantyID); business rule; product can have up to one warranty*

PRODUCT_WAREHOUSE
Junction relation for PRODUCT stored in WAREHOUSE (models many-to-many if applicable)

Attribute	*Data Type*	*Constraint / Notes*
ProductID	*CHAR(12)*	*Primary Key (composite): FK references PRODUCT (ProductID)*
WarehouseID	*CHAR(8)*	*Primary Key (composite); FK references WAREHOUSE (WarehouseID)*
QuantityOnHand	*NUMBER(10)*	*NOT NULL; CHECK (QuantityOnHand >=0)*

REPAIR

Relation for the entity REPAIR

Attribute	*Data Type*	*Constraint / Notes*
RepairID	*CHAR(12)*	*Primary Key; NOT NULL*
ServiceCenterID	*CHAR(8)*	*FK references SERVICE_CENTER(ServiceCenterID); NOT NULL*
ProductID	*CHAR(12)*	*FK references PRODUCT(ProductID); NOT NULL*
TechnicianID	*CHAR(10)*	*FK references EMPLOYEE(EmployeeID); NULLABLE*
DateofRequest	*DATE*	*NOT NULL*
Reasoning	*VARCHAR(500)*	*NULLABLE*
Repair Status	*VARCHAR(30)*	*NOT NULL; CHECK (RepairStatus IN ('Pending', 'InProgress','Completed','Returned'))*

REPAIR_PARTS
Junction relation linking REPAIR to REPLACEMENT_PARTS (many parts per repair)

Attribute	*Data Type*	*Constraint / Notes*
RepairID	*CHAR(12)*	*Primary Key (composite); FK references REPAIR(RepairID)*
PartID	*CHAR(12)*	*Primary Key (composite); FK references REPLACEMENT_PARTS*

		(PartID)
QuantityUsed	*NUMBER(6)*	*NOT NULL; CHECK (QuantityUsed > 0)*

REPLACE
Relation for the entity REPLACE (replacement actions)

Attribute	*Data Type*	*Constraint / Notes*
ReplaceID	*CHAR(12)*	*Primary Key; NOT NULL*
ServiceCenterID	*CHAR(8)*	*FK references SERVICE_CENTER(ServiceCenterID); NOT NULL*
ProductID	*CHAR(12)*	*FK references PRODUCT(ProductID); NOT NULL*
TechnicianID	*CHAR(10)*	*FK references EMPLOYEE(EmployeeID); NULLABLE*
DateOfRequest	*DATE*	*NOT NULL*
Reasoning	*VARCHAR(500)*	*NULLABLE*
ReplacementPartID	*CHAR(12)*	*FK (nullable) references REPLACEMENT_PARTS(PartID)*
ReplaceStatus	*VARCHAR(30)*	*NOT NULL; CHECK (ReplaceStatus IN ('Pending', 'Completed', 'Returned))*

REPLACEMENT_PARTS
Relation for the entity REPLACEMENT_PARTS

Attribute	*Data Type*	*Constraint / Notes*
PartID	*CHAR(12)*	*Primary Key; NOT NULL*
PartName	*VARCHAR(100)*	*NOT NULL*
Category	*VARCHAR(50)*	*NULLABLE*
PartDescription	*VARCHAR(500)*	*NULLABLE*
Availability	*NUMBER(6)*	*NOT NULL; CHECK (Availability >=0)*
WarehouseID	*CHAR(8)*	*FK (nullable) references WAREHOUSE(WarehouseID)*
ServiceCenterID	*CHAR(8)*	*FK (nullable) references SERVICE_CENTER(ServiceCenterID)*

RETURN
Relation for the entity RETURN

Attribute	*Date Type*	*Constraint / Notes*
ReturnID	*CHAR(12)*	*Primary Key; NOT NULL*
CustomerID	*CHAR(10)*	*FK references CUSTOMERS(CustomerID); NOT NULL*
ProductID	*CHAR(12)*	*FK references PRODUCT(ProductID); NOT NULL*
WarehouseID	*CHAR(8)*	*FK (nullable) references WAREHOUSE (WarehouseID); NULLABLE*
ReturnDate	*DATE*	*NOT NULL*
Reason	*VARCHAR(500)*	*NULLABLE*

ItemCondition	*VARCHAR(30)*	*NOT NULL; CHECK (ItemCondition IN ('New'.'LikeNew', 'Used', 'Damaged'))*
ReturnStatus	*VARCHAR(30)*	*NOT NULL; CHECK (ReturnStatus IN ('Received','Inspected', 'Refunded', 'Repaired','Replaced'))*

SERVICE CENTER
Relation for the entity SERVICE_CENTER

Attribute	*Data Type*	*Constraint / Notes*
ServiceCenterID	*CHAR(8)*	*Primary Key; NOT NULL*
Name	*VARCHAR(100)*	*NOT NULL*
Address	*VARCHAR(200)*	*NOT NULL*
Capacity	*NUMBER(6)*	*NULLABLE;CHECK (Capacity >=0)*
ContactInfo	*VARCHAR(100)*	*NULLABLE*
WorkforceNotes	*VARCHAR(200)*	*NULLABLE*

SERVICE_REPORT
Relation for the entity SERVICE_REPORT

Attribute	*Data Type*	*Constraint / Notes*
SReportID	*CHAR(12)*	*Primary Key; NOT NULL*

ServiceRequestID	*CHAR(12)*	*FK (nullable) references SERVICE_REQUEST(Ticket Number)*
HubID	*CHAR(8)*	*FK (nullable) references DELIVERY_HUB(HubID)*
AccountID	*CHAR(10)*	*FK (nullable) references SONY_ACCOUNT(AccountID)*
DateofCreation	*DATE*	*NOT NULL*
IssueResolved	*VARCHAR(10)*	*NOT NULL; CHECK (IssueResolved IN ('Yes','No'))*
Location	*VARCHAR(200)*	*NULLABLE*
Review	*VARCHAR(1000)*	*NULLABLE*

SERVICE_REQUEST
Relation for the entity SERVICE_REQUEST

Attribute	*Data Type*	*Constraint / Notes*
TicketNumber	*CHAR(12)*	*Primary Key; NOT NULL*
CustomerID	*CHAR(10)*	*FK references CUSTOMERS(CustomerID); NOT NULL*
ConsultantEmployeeID	*CHAR(10)*	*FK (nullable) references EMPLOYEE(EmployeeID)*
DateofRequest	*DATE*	*NOT NULL*
IssueDescription	*VARCHAR(1000)*	*NULLABKE*
Status	*VARCHAR(30)*	*NOT NULL; CHECK (Status IN ('Open', 'Assigned', 'InProgress', 'Closed'))*

SERVICE_REVIEW
Relation for the entity SERVICE_REVIEW

Attribute	*Data Type*	*Constraint / Notes*
ReviewID	*CHAR(12)*	*Primary Key; NOT NULL*
CustomerID	*CHAR(10)*	*FK references CUSTOMERS(CustomerID); NOT NULL*
ServiceRequestID	*CHAR(12)*	*FK (nullable) references SERVICE_REQUEST(Ticket Number)*
ProductID	*CHAR(12)*	*FK (nullable) references PRODUCT(ProductID)*
Rating	*NUMBER(1)*	*NOT NULL; CHECK (Rating BETWEEN 1 AND 5)*
Comments	*VARCHAR(1000)*	*NULLABLE*
Media	*VARCHAR(200)*	*NULLABLE (link)*
DateOfCreation	*DATE*	*NOT NULL*

SONY_ACCOUNT
Relation for the entity SONY_ACCOUNT

Attribute	*Data Type*	*Constraint / Notes*
AccountID	*CHAR(10)*	*Primary Key; NOT NULL*
AccountCreatedDate	*DATE*	*NOT NULL*
AccountStatus	*VARCHAR(20)*	*NOT NULL; CHECK (AccountStatus IN*

		('Active','Suspended','Closed '))
AccountType	*VARCHAR(30)*	*NULLABLE*
CustomerID	*CHAR(10)*	*FK references CUSTOMERS(CustomerID); NOT NULL*
EmailOnFile	*VARCHAR(100)*	*NULLABLE*

STORE
Relation for the entity STORE

Attribute	*Data Type*	*Constraint / Notes*
StoreID	*CHAR(8)*	*Primary Key; NOT NULL*
Name	*VARCHAR(100)*	*NOT NULL*
Address	*VARCHAR(200)*	*NOT NULL*
StoreType	*VARCHAR(50)*	*NULLABLE*
ContactInfo	*VARCHAR(100)*	*NULLABLE*

WARRANTY
Relationship for the entity WARRANTY

Attribute	*Data Type*	*Constraint / Notes*
WarrantyID	*CHAR(10)*	*Primary Key; NOT NULL*
OrderDetailedID	*CHAR(12)*	*FK (unique) references ORDER_DETAILS*

		(OrderDetailID); NULLABLE; enforce 1:1 by UNIQUE (OrderDetailedID)
StartDate	*DATE*	*NOT NULL*
DateOfExperation	*DATE*	*NOT NULL; CHECK (DateOfExperation >= StartDate)*
Status	*VARCHAR(20)*	*NOT NULL; CHECK (Status IN (‘Active’. ‘Expired’, ‘Voided))*
CoverageType	*VARCHAR(50)*	*NULLABLE*

WAREHOUSE
Relation for the entity WAREHOUSE

Attribute	*Data Type*	*Constraint / Notes*
WarehouseID	*CHAR(8)*	*Primary Key; NOT NULL*
Address	*VARCHAR(200)*	*NOT NULL*
Capacity	*NUMBER(10)*	*NULLABLE; CHECK (Capacity >= 0)*
ContactInfo	*VARCHAR(100)*	*NULLABLE*
OperatingHours	*VARCHAR(100)*	*NOT NULL; default ‘24/7’*

ER → Relational Notes

1. Superclass / Subclass (EMPLOYEE): Implemented as class-table inheritance: EMPLOYEE holds common attributes; TECHNICIAN and CLIENT_SERVICE_CONSULTANT are separate tables with EmployeeID as PK and FK to EMPLOYEE. This enforces 1:1 subclass membership and keeps data normalized.

2. WARRANTY ←→ ORDER_DETAILS: modeled as a separate WARRANTY table with OrderedDetailedID as a unique FK to ORDER_DETAILS to enforce the 1:1 relationship. OrderDetailedID may be NULL in WARRANTY if some warranties are not tied to an order detail (adjust if business rule requires otherwise).

3. CUSTOMERS ←→ SONY_ACCOUNT: SONY_ACCOUNT. CustomerID is NOT NULL (each account belongs to a customer). The ER states a person may have multiple accounts, and at least one must be recorded. Enforcing "each customer must have at least one account" is best done in application logic or via a periodic DB constraint (e.g., trigger or deferred check).

4. PRODUCT STORAGE: ER suggests products may be stored in warehouses and sold in stores. If a product can be in multiple warehouses, the PRODUCT_WAREHOUSE junction table models that PRODUCT.StoreID is nullable to allow online-only products.

5. DELIVERY_HUB: deliveryType constrained to ('Ground","Air'). For shipment tracking, consider adding a SHIPMENT table if you need to track individual deliveries (HubID, ProductID, CustomerID, ShipmentDate, DeliveryStatus).

6. REPAIR / REPLACE AND PARTS: REPAIR_PARTS junction table models many parts per pair. REPLACE references REPLACEMENT_PARTS when replacement uses a part; if replacement is a whole product, Replace. ReplacementPartID may be NULL and Replace. ProductID used.

7. RETURN: Return.WarehouseID is nullable because returns may be processed at stores or warehouses. ItemCondition and ReturnStatus have CHECK enumerations.
8. CONSTRAINTS ENFORCEMENT: some business rules (e.g., "each customer must have at least one account") are difficult to enforce purely with standard FK/NOT NULL constraints: use triggers or application-level checks where necessary/